

Large Ping Rocks Set-Up

I needed to consolidate the number of setups I was caring for, so I decided to amass as much of my Ping collection as possible onto 3 large lava rocks to all fit in one giant fancy reservoir. I chose 3 large instead of one huge rock so I could pick them up and move them around when necessary. Also, 3 large instead of one huge actually gives more surface area to work with, and I had a lot of Pings! (They're just like Tribbles...) This project required a kind of "playing the long game" mentality and took several months to complete from start to finish. And it's best to have lots of spare parts on hand like mossy bits, fill-in Utrics, etc.

First, I undid my pots and placed the Pings and Utrics on wet paper towels or sand on cookie sheets & plant trays. This helped them go dormant, which is preferable for handling and planting. I also kept watering the leftover pots of mossy bits to keep them green to re-purpose for use in the project. I sourced rocks that would be tall enough to show the plants above the rim of the large square enclosure I planned to use and included an egg crate that I cable tied on a snap-together raised gravel bed for better water flow under the rocks. (Aquarium supplies are a great source for these.)



A few unpotted bits & pieces

If you source your lava from an outside landscaping place like I did and not from someplace that has already cleaned the rocks, wash and scrub them down, pick out the dried bits of plant matter, etc., to minimize unwanted insect eggs from hatching, or fungus and weeds from popping up to destroy your little paradise. Scrub the rocks a second time if you notice anything suspect. I even sprayed them with CP safe insect repellent a couple of weeks before beginning.



And because I enjoy finding ways to push the boundaries of what's possible, I also wanted to include some Drosera on these rocks. So, I needed to find a method of creating steady moisture levels for plants with differing water needs on rocks that had their own individual degrees of natural porosity, and I wanted to do it in a way that would create a low maintenance display. Enter wicking string!

Wicking string, like dry sphagnum is hydrophobic until thoroughly wet, so you always want to work with sopping wet wicking string. (I found this out the hard way. I should have just held it under water and squeezed repeatedly until saturated.) You'll want to place your rock in a tub with the depth of water that sorta matches the level of your future



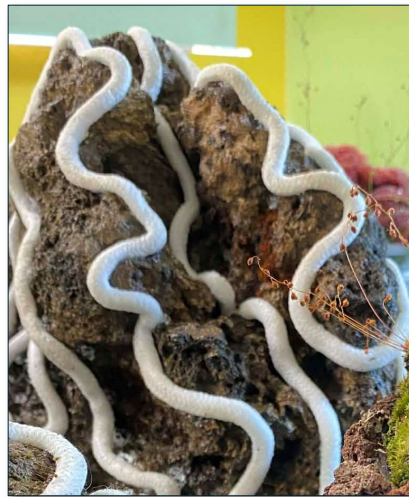
display. This allows the string to dangle in the water while gluing it on and keeps the visual of approximately where the water line will be in your design. Super glue gel works fine for this. Just remember to always wear goggles and a mask when working up close for long periods of time with this glue. It will stick to your fingers, so keep wet paper towels handy. And it will rip your nasal passages to shreds and destroy the moisture balance in your eyes for DAYS if you just go in unprotected. (Ask me how I know...)

Another thing about the rocks – sometimes they wobble. If you want to use a rock that wobbles, you will need to stabilize it. Once it's full of water and brimming with fragile gorgeousness, you will still need to be able to move it around and pick it up while minimizing fear of tipping. So, if like most people you don't have an appropriate saw to shave off the bumpy bits on the bottom, you can glue some small stabilizing support rocks to act like a shim. Give it a day to verify it's stabilized. Add more "shim" rocks as needed. I keep a bag of fire pit lava rocks around for stuff like this. Also, bear in mind the addition of string, mossy bits and big adult Pings on the rocks will increase its mass and thus the space you need around the rocks, so if using multiple rocks, leave more room than you think you will need between the rocks and to the edges of the reservoir enclosure as well.



Once the rocks are clean, stabilized, and dry, soak them in a few inches of water for a day to see how much natural absorption they each have and where the drier areas are. Since wicking defies gravity, I determined that these less porous areas of the rocks would benefit from shorter paths between the string dipping back into the water under the rock. This creates easier capillary action for the rock's drier areas. So, I glued the string path up to a dry area and glued it back down again to dip back into the water without meandering the string to yet another area first. The pattern ended up creating a visual similar to worm canals. But worms have to keep themselves moist too, so maybe that makes some weird kind of sense.

Don't rush this wicking process. This is the infrastructure that makes everything else so much easier for the life of the display. I planned the deeper crevasses in the rocks to house the Drosera and made sure it was either naturally damp, or I placed wicking string down into those areas and around their borders to provide water laterally and for their roots as they grow into the rock. I also strategically placed string along the vertical sides of rocks to create little shelves for Pings to hang onto.



On the bottom of rocks where the string will be underwater, glue the wicking string into little fissures to help hide it from view. Make sure your string path dips down into the water periodically underneath the rock, and that you leave plenty of string in those hidden underwater portions of your rocks to ensure they always have plenty of soaking string to maintain optimum capillary action to the entire display. Once you cover it with plants, no one will know your wicking secret unless you tell them.





One other note about string placement. Notice where you naturally want to pick up each rock. Prepare the entire display to allow you to be able to lift that rock up from some really stable places along the bottom. Don't glue string there. Don't plant anything near there. Plan to be able to take that display apart and clean it without disturbing your masterpiece, keeping in mind that young or dormant Pings will double or even triple in size! Flushing your reservoir regularly keeps TDS low and maximizes plant health.

Once you've got all the wicking string in place, let the pieces sit in their reservoir of water for a few days. Then check the string at the point of the rocks that were dry and the highest points as well to verify that they are staying damp.

Next, it's time to prepare your work area for the rest of the project, as well as set up your media and gather your plants and mossy bits. I created a finely strained peat & sand combo using a strainer to sift out the chunkier peat bits, then mixed with maybe 1/3 sand and added enough water to make a paste. This acts as a natural glue when needed to camouflage string and to provide security or added moisture to a plant. Save your mossy bits to glue on the undersides of string still showing after plants are in place (because media doesn't glue well upside down). I added various Utrics colonies as ground cover.



Set your work area up with a clamp-on desk lamp on an arm that you can bring down for close-up work, wear a head lamp if you have one, include an area for trash, a bowl of water for the constant tool and hand cleaning you will do, paper towels, goggles (or



readers if you're like me and can't see close up so good), a mask, and a good selection of pointed tweezers, tweezers with an angled tip, long tweezers, a couple of spoons, tiny crafting scissors, etc., and place everything on large towels or some other surface you can use to gather the detritus for easy cleanup. And buy the small Super Glue gel tubes in bulk. Larger ones are hard to manage one-handed and get clumpy & gross. The glue likes to come out all at once sometimes, so be prepared for that, and don't let it make you feel rushed. (I had to remind myself repeatedly that the glue was not the boss of me!) Keep a couple of paper towels wet to wipe glue nozzles and to help keep tools clean as you go.





The planting should be done in phases if you're planning larger pieces. I took some time to determine a design layout that grouped species together next to off-setting colors for contrast. I also took advantage of certain angles and surfaces to maximize larger groups of one species or to highlight a particularly vibrant combination of plants. Once you have the elements of design figured out, you need to break up the planting process by where a specific grouping will be on the rock, how long that area will take to plant, and what would be the best order of business, so you don't knock around previous plantings. For example, I knew I wanted the *Drosera* species along the crevasse or in holes on the rocks, and they would be easier to plant before adding any Pings to the rock, not only because of the actual planting process, but because I wanted to give them time to show me they were getting enough water before loading the rocks with Pings.

Once I knew the *Drosera* were getting enough to drink, I started gluing Pings along my pre-designated pattern. Yes, I did just use the Super Glue gel in many cases to stick them on, especially if it was directly on a shelf created by wicking string. I also used the pre-mixed media in places that seemed a bit drier and that might need the additional wicking of the peat moss. If you have a limited supply of actual mossy bits & Utrics scavenged from abandoned pots, wait until you're done planting Pings to hide the underside of remaining exposed string with mossy bits. I had a plethora of bits and Utrics, so I used them as I went along. Ration according to your supply. Or grow some extra intentionally from what you have in the planning stages of the project and by the time you get to the planting, you'll probably have a nice amount.



The rest of the project is up to your inventory on hand and creative inspiration. Some tips on placement – if you have *P. emarginata* or the hybrids, they can handle a bit more water, so closer to the bottom or in wetter areas of the rock would be good. Same with any special Utrics you care about seeing on the regular. (Utrics will reach for the water and create a cascade effect if you like that look.)





If something likes it a bit drier, like a *P. gypsicola* hybrid, consider a drier spot where you can also install a finely ground mineral mix. If there is a species you want to highlight, design it as a centerpiece or a top of the mountain eye-catcher. Bold color contrasts highlight both types of plants better than blending them randomly. Plant some species in a

line, and others in a zigzag, or a circle. Follow the line of part of the rock around with a species for sweeping effects. Fill in spaces between several groups of larger species with colonies of a smaller one. Keep your lighting close to the display for best color.



I used Yescom white panels, so any similar LED panel works fine. The display was also located just inside a southwest facing glass door, so it got some afternoon sun.

I randomly fertilized with a light misting of Schultz Cactus Fertilizer (14 drops in 1 gallon RO/distilled water.) To keep it clean, use an aquarium vacuum siphon and bucket to drain the water every few weeks, then pour fresh RO to stir up more sediment and siphon it a second time. Do this until it's clean to your satisfaction. Most of the time, you will only need to lift the rocks out if it gets too slimy to do just the siphon method. Then you would need to wash the enclosure and wipe the slime off the bottom parts of the rocks, but a regular flushing routine can help keep it from getting that way.

The structure I housed this display on is a heavy duty tiered adjustable metal shelving unit. Love those things. We used hooks to hang the light panels from the wire shelf above the display, and placed wood planks below the reservoir which we had on a turn table. That way the whole display can spin with a simple push for easy trimming and gawking.

One last tip – make sure before you plant that you can easily remove the rocks without running into the top of the shelf and/or the lights, since you will have to lift the rock straight up and pull it to you. You won't easily be able to slightly tip it, turn it, etc. as it will be covered in floppy, delicate, wet life forms. And keep in mind, once the plants get to their mature size you will need even more space to do this maneuver, and it will be heavier. Or you can construct something that moves the entire structure above the display out of your way. Knowledge is power.

